Independence Assessment

Overall

StreetLight volume estimates and Iowa DOT counts should align closely because both systems measure the same underlying traffic volumes. High agreement is therefore expected but does not, by itself, imply that the two data sources are methodologically dependent.

The relevant concern is not agreement, but whether Iowa DOT counts are used to directly produce StreetLight estimates (what I will henceforth refer to as "methodological independence"). Based on available StreetLight documentation, FHWA guidance, and independent state-level evaluations (including Minnesota DOT), current evidence supports StreetLight estimates being primarily probe-driven, with DOT counts used for calibration and validation, rather than for direct substitution or enforcement of agreement at specific locations.

To emphasize the distinction: Calibration can improve model performance without compromising methodological independence, provided that evaluation is conducted on held-out data. As a matter of due diligence, however, confirmation of methodological independence would benefit from explicit vendor clarification that Iowa DOT count data are not used as direct inputs to model training or segment-level estimation, but are limited to calibration and validation, and that the StreetLight methodology does not incorporate Iowa DOT data in a manner that would directly constrain estimated volumes.

Summary

Based on publicly available documentation and independent evaluations:

- StreetLight estimates are derived primarily from probe/mobile/GPS data, not directly from DOT traffic counters (StreetLight documentation)

- Traffic counts are used mainly for model calibration and accuracy assessment, consistent with standard traffic modeling practice (StreetLight documentation; FHWA guidance)

- Out-of-sample (holdout) testing is explicitly reported in validation studies, which reduces the risk of circularity (StreetLight documentation)

- The primary unresolved issue is the *extent of Iowa-specific count data used during calibration*, which represents the main remaining risk to full independence

Implications

- The principal risk to monitor is *local calibration or segment-level adjustment using Iowa DOT counters*
- Strong agreement between DOT counts and StreetLight estimates is expected, given that both estimate "true traffic volume"

- Correlation or low error alone cannot diagnose methodological dependence (statistical tests of independence are limited in scope)

- Definitive confirmation of methodological independence requires vendor disclosure or Iowa-specific hold-out testing, not additional statistical comparisons alone

Taken together, current evidence supports interpreting StreetLight volumes as independently generated but empirically calibrated, consistent with FHWA-recommended validation practice.

More Details

The following provides further details on basis for this assessment, i.e., why agreement is expected, how independence should be evaluated, what evidence is currently available, and what uncertainties remain.

Why agreement is expected

When comparing StreetLight estimates to Iowa DOT or JACKALOPE observed counts, high correlation is inevitable because both aim to measure the same latent quantity: True average annual traffic volume.

- High correlation is therefore necessary but not diagnostic.
- The relevant question is not *whether* the estimates agree, but *why* they agree.

What methodological independence means in practice

Methodological independence depends on *how counts enter the modeling pipeline*, not whether they appear anywhere in it:

- Acceptable: Counts used for calibration and model tuning, with performance evaluated on held-out sites (FHWA guidance)

- Problematic: Counts directly constraining, replacing, or being used as ann input for estimating specific segments

The practical question is:

> To what extent are Iowa DOT counts used to *produce* StreetLight Iowa estimates, rather than simply evaluate them?

Current evidence indicates that Iowa DOT counts are used for calibration and validation rather than direct estimation, though documentation does not fully rule out the use of Iowa-specific count data during model training.